Karan Chitkara

701-200-1689 | karan.chitkara@outlook.com | LinkedIn | Google Scholar

SUMMARY

Senior Principal Software Engineer with 12+ years of experience building safety-critical medical device software for neurostimulation therapy. Inventor on 5 U.S./international patents (3 granted, 2 pending). Software Development Lead for the Medtronic Percept system, recognized as a TIME Best Invention of 2025. Recipient of Medtronic's Technical Contributor of the Year Award (top <1% of 95,000+ employees). Deep expertise in mobile/embedded architectures, cryptography, automation, and regulatory compliance across FDA, TÜV, and NMPA.

TECHNICAL SKILLS

Languages	C, C++, C#, Swift, Kotlin, Java, JavaScript, Python, SQL
Platforms	iOS, Android, Kotlin Multiplatform (KMM), Bluetooth Low Energy (BLE)
Tools	Xcode, Android Studio, IntelliJ, Visual Studio, Git, Jenkins
Architecture	MVVM, MVP, MVC, FLUX, Dependency Injection, Modular Frameworks
Domains	FDA/TÜV/NMPA Compliance, HIPAA/GDPR, Cryptography, Medical Device Software (IEC 62304)

PROFESSIONAL EXPERIENCE

Senior Principal Software Engineer — Medtronic plc — Jun 2025 – Present

- Define technical roadmap for next-generation External Neurostimulator (ENS) platform, including modular architecture, telemetry acceleration, and security-first design to replace legacy systems globally.
- Lead cross-functional alignment across human factors, systems, clinical, and firmware teams to drive end-to-end design traceability and regulatory readiness.
- Architect multi-device reusability frameworks and dynamic signal-driven programming systems, delivering strategic IP contributions across multiple patent filings.
- Establish enterprise-level software governance — static/dynamic analysis protocols, code health infrastructure, and scalable design patterns for safety-critical systems.

Principal Software Engineer — Medtronic plc — Dec 2021 – Jun 2025

- Architected scalable software frameworks for Deep Brain Stimulation (DBS) systems, leading cross-platform mobile architectures (iOS & Android) adopted by multiple therapy teams.
- Led the Electrode Identifier feature — automating electrode selection and titration — recognized as a TIME Best Invention of 2025, saving clinicians hours per programming session.
- Built the Automation Runner Tool, analyzing 17M+ lines of code and executing 20,000+ test cases, reducing manual validation effort by 90% and accelerating regulatory approval timelines from months to days.
- Shipped major DBS releases (Sensight Directional Lead, Adaptive DBS) across FDA, TÜV, and NMPA markets.
- Designed proprietary cryptographic frameworks for BLE-based neurostimulator communication, ensuring HIPAA/GDPR-compliant patient data protection.

Senior Software Engineer — Medtronic plc — Jul 2018 – Dec 2021

- Designed and shipped a cross-therapy mobile communication platform (Android & iOS) now adopted by multiple therapy teams across Medtronic's Neuromodulation division, eliminating redundant development efforts.
- Implemented Kotlin Multiplatform (KMM) framework for shared business logic between iOS and Android, reducing code duplication and accelerating feature delivery.
- Delivered reusable SDKs with documentation, dependency injection, and automated test frameworks, cutting onboarding time for new therapy teams.

Software Engineer — Medtronic plc — Apr 2015 – Jun 2018

- Developed Android applications for DBS therapy with secure BLE communication protocols, enabling real-time clinician-device interaction.
- Built automated UI screen-capture and validation tooling, reducing manual regulatory compliance effort.
- Resolved critical software defects and implemented modular design patterns (MVC, MVP, FLUX), improving long-term maintainability.

Software Developer — Open Access Technology Inc. — Jan 2014 – Mar 2015

- Resolved 150+ software defects and developed new modules (C#, JavaScript, SQL Server) for enterprise power trading and energy generation platforms.
- Built C++-based optimization algorithms for energy unit commitment and load-balancing across regions.

Java Programmer — North Dakota State University Apr 2012 – Dec 2013

- Built a full-stack web application (Java, MySQL, HTML/CSS/JS) to digitize the university's course proposal approval process, eliminating paper-based workflows.

PATENTS

GRANTED **US12144637B2** — Brain Stimulation and Sensing: real-time therapy delivery optimization and brain signal monitoring

GRANTED **US11986663B2** — Offline Session Review & Interactive Programming: clinician data review without device connectivity

GRANTED **US11571576B2** — Brain Stimulation and Sensing: automated signal processing for adaptive DBS therapy

PENDING **WO2025029593A1** — UI for Electrode Sensing Results in a Medical Device System

PENDING **WO2025029591A1** — Electrode Ranking and Configuration Scoring for Medical Device Programming

AWARDS

Medtronic Technical Contributor of the Year (2024) — Awarded to <1% of 95,000+ global workforce for leading the Electrode Identifier, a breakthrough in therapy automation.

TIME Best Inventions 2025 — Software Development Lead for the Medtronic Percept Deep Brain Stimulation system.

Medtronic Sensight Award — Recognized for delivering the Sensight Directional Lead System across multiple international markets.

EDUCATION

Master of Science, Software Engineering — North Dakota State University, Fargo, ND Dec 2014

Bachelor of Science, Computer Science — North Dakota State University, Fargo, ND May 2012